

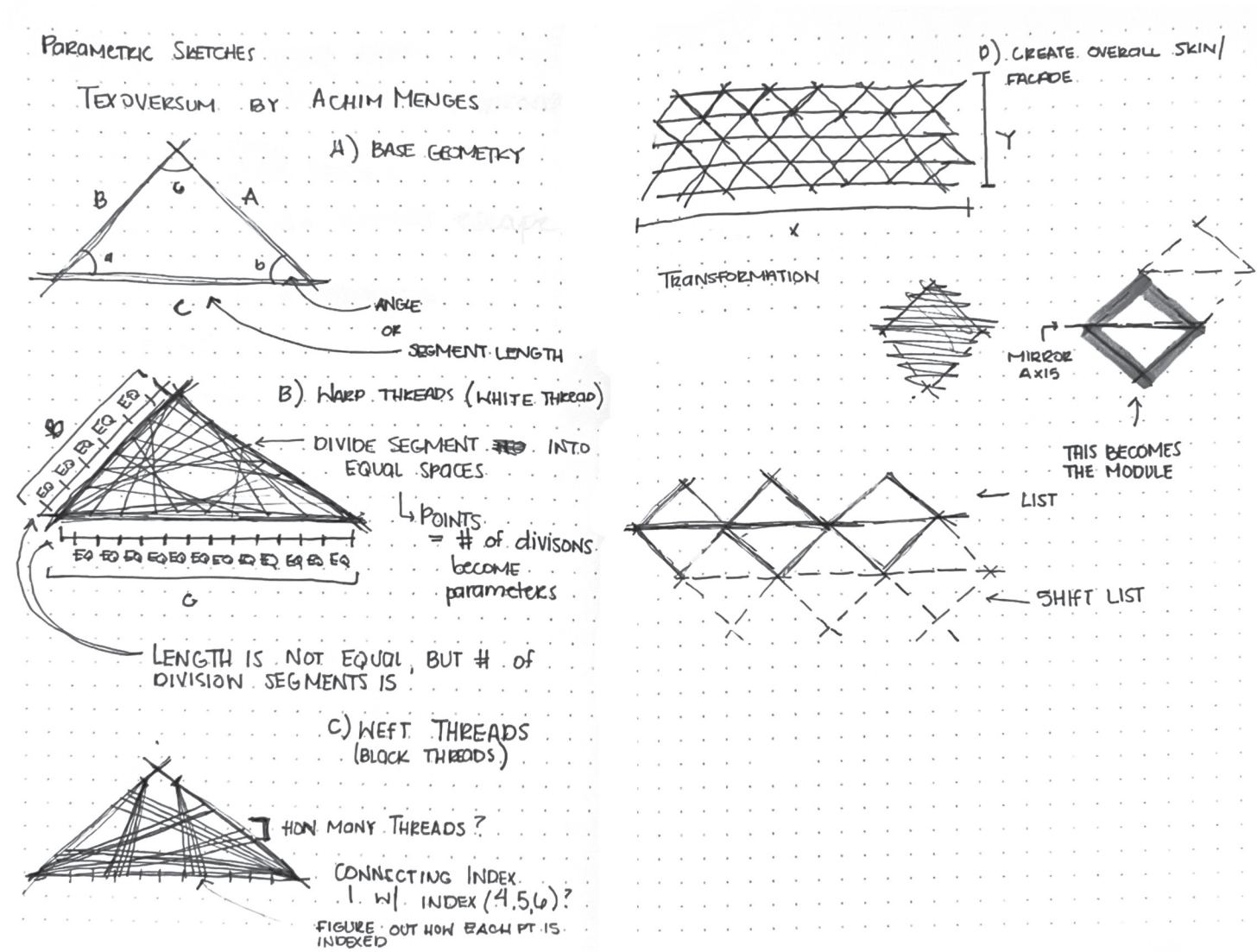
ASSIGNMENT 2 // PAVILLION

For my master’s studio project, I have been studying the logic and form generated through weaving. Because my research has been largely through analogue mediums, I wanted to use this assignment as an opportunity to study weaving through a digital lens. Via in class examples and personal research, the project Texoversum by Allmannwappner + Menges Scheffler Architekten + Jan Knippers Ingenieure provided the main inspiration for this assignment and the resulting pavillion.

INSPIRATION



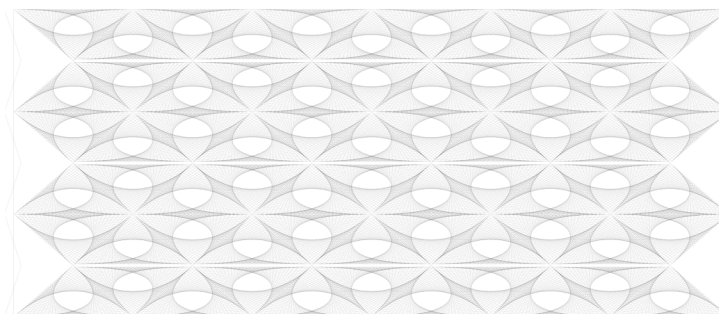
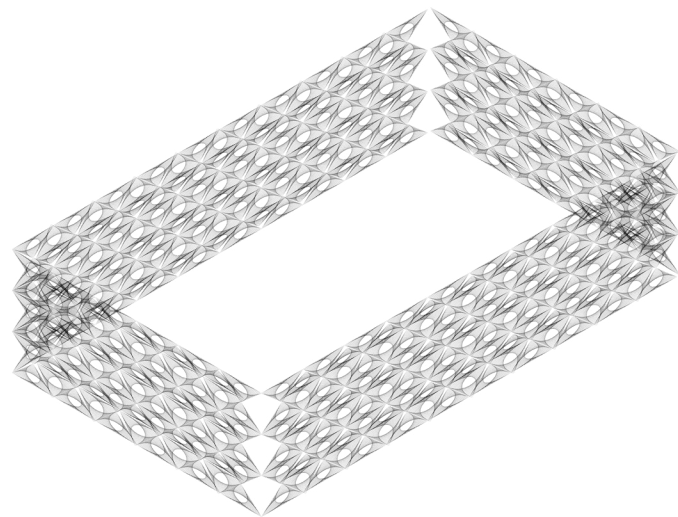
SKETCHES



The series of sketches shown above identifies the parameters of the assembly in an effort to formulate a plan before transitioning into rhino + grasshopper. The first parameter identified are the angles and the lengths of the triangles, before even considering the threads themselves. The white threads in the facade act as the warp and the black threads act as the weft resulting in 2 different parameters. After the individual units are made, the third parameter is how the modules are arranged to create the pavillion.

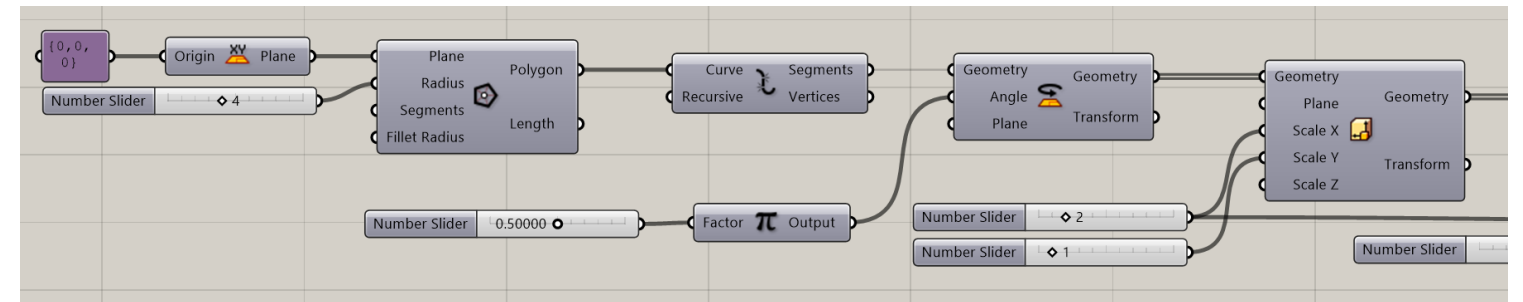
GRASSHOPPER WORKFLOW // FIRST DRAFT

The first script that I worked on followed a tutorial I found online for generating the individual units that I then manipulated to create the resulting form. While this attempt was close to the inspiration, all of the transformations applied to the units made the logic convoluted and it became difficult to address gaps in the facade. While I did not use this script for my final pavillion design, I learned that the paramteres I identified in my sketches can be applied in any order. In this case specifically, it became helpful for me to consider building the structure and then populating the units.

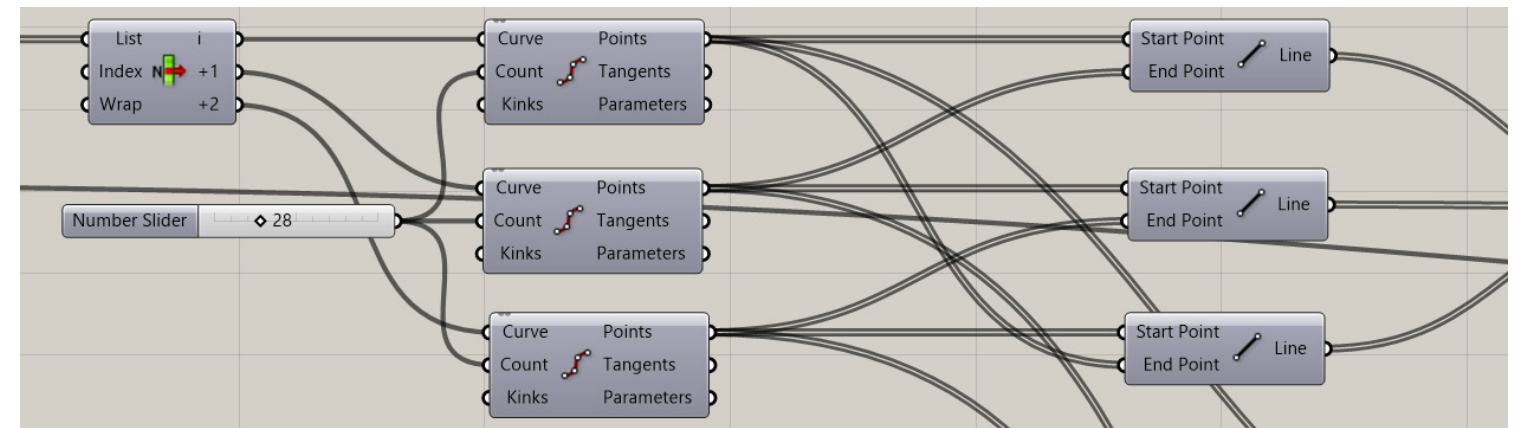


DRAFT 1 GENERAL OUTLINE

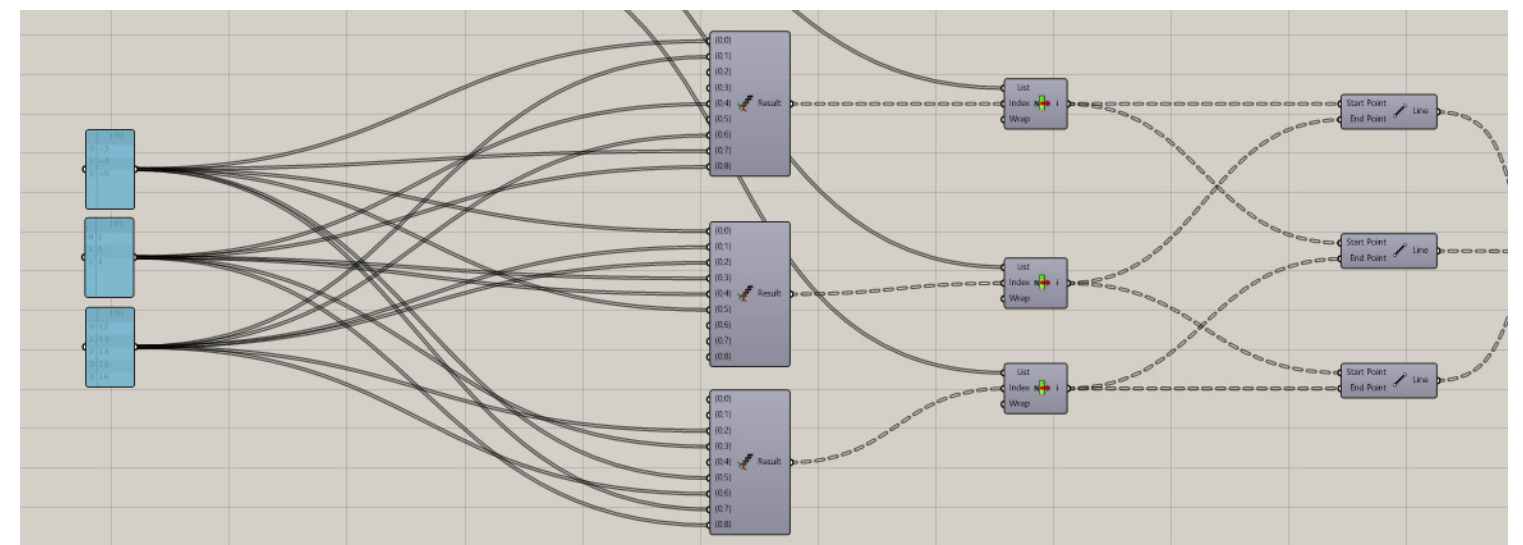
STEP 1 Creating base geometry: acute triangles.



STEP 2 Making the white threads: *The Warp*

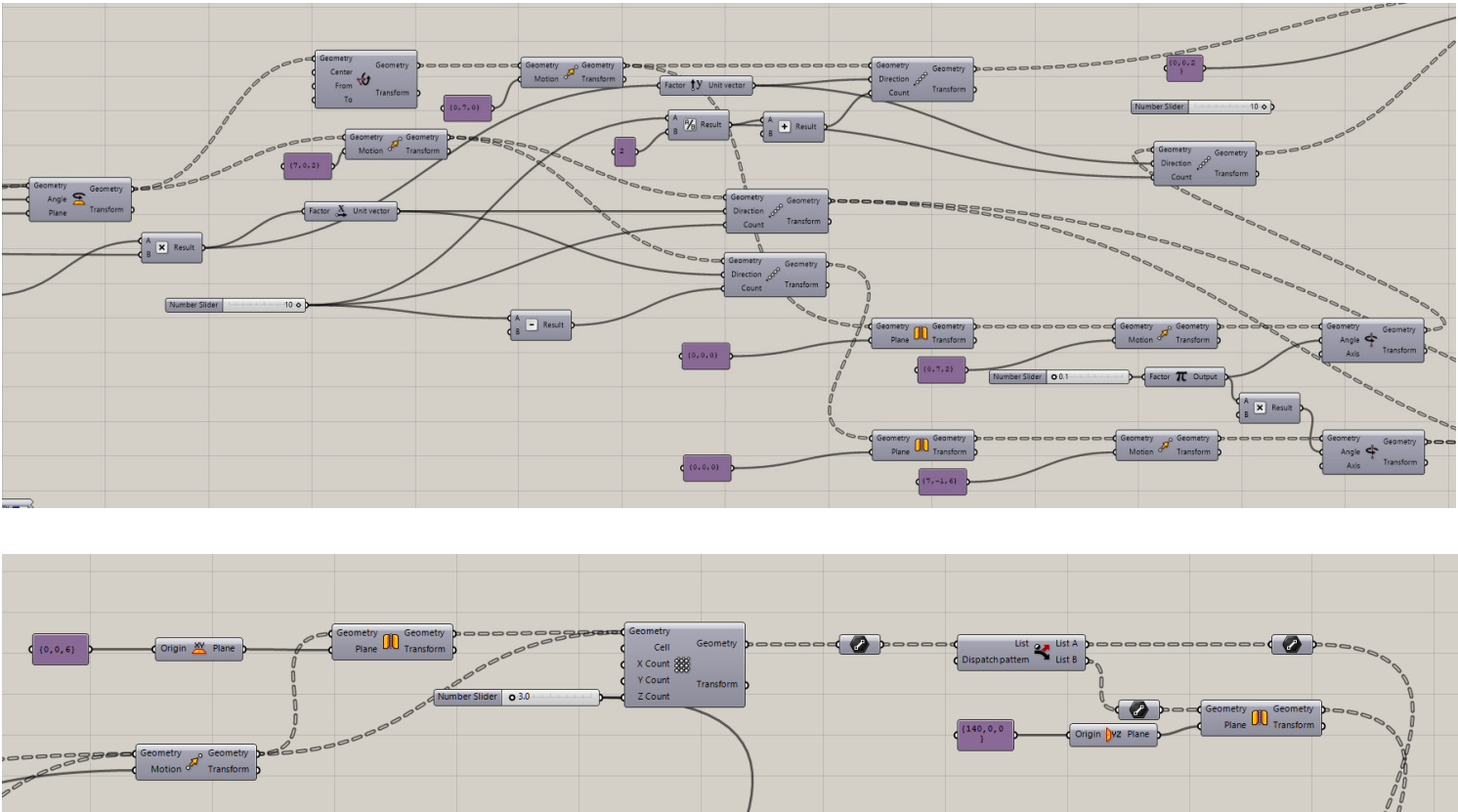


STEP 2 Making the black threads: *The Weft*



DRAFT 1 GENERAL OUTLINE continued

STEP 4 Transforming the modules



After writing this script, I found that the large majority of time and effort was spent on arraying the modules and manipulating them in a way that felt almost manual. As a result, it felt that this script was not an efficnet use of grasshopper and its utility in computational design.

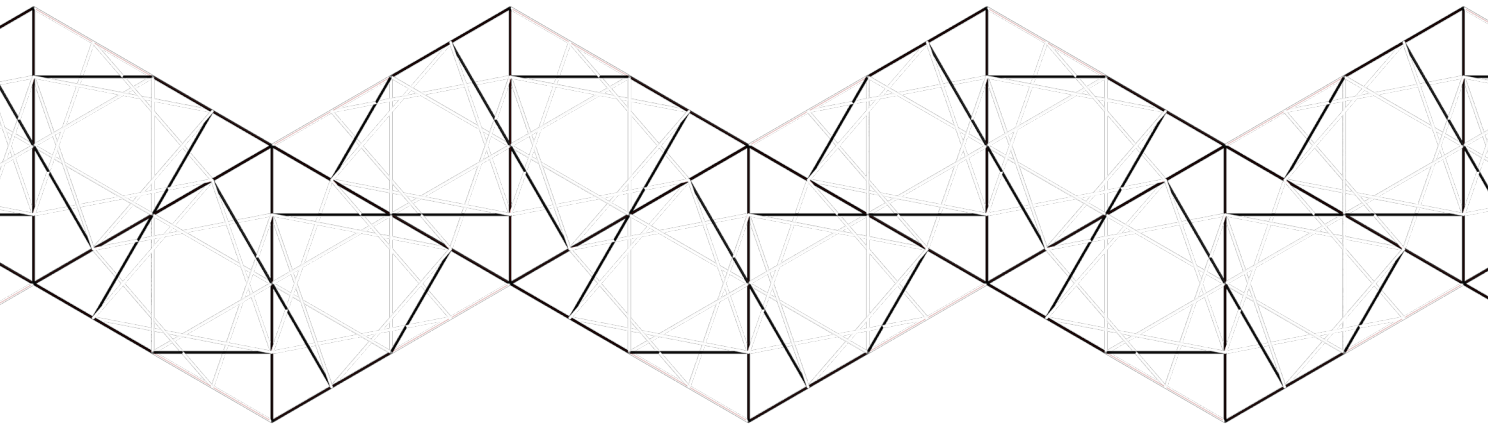
After consulting Chris for some help, it seemd like the best course of action would be to simplify my logic greatly and approach this problem in the complete opposite way as I had initially thought about it. Using a base script for reference my new plan became:

- STEP 1 Make the Frames (structure)
- STEP 2 Transform the Frames
- STEP 3 Weave the Threads

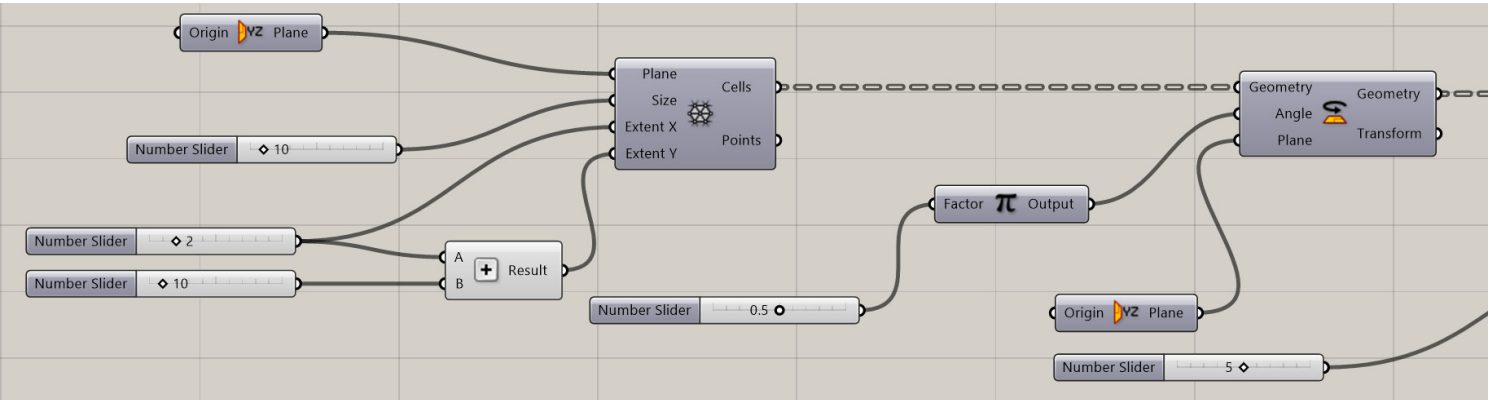
The frames are generated using the triamgular component which creates a triangular grid. I then rotated the triangular grid on the YZ plane so I could get the desired form for the “weaving” to occupy.

GRASSHOPPER WORKFLOW // FINAL

As previously mentioned, this script was based off of a script provided to me. Instead of focusing on generating the individual modules and moving them, the first step streamlines the geometry generation of the structure to be a series of modules arrayed in a grid formation and then translated. From there, a similar process for creating the warp is used. To create the black threads, a similar process to the formation of the white threads is used, resulting in a more efficient script. The primary benefit of reworking this script is that the geometry for the frames and threads is retained as individual elements.



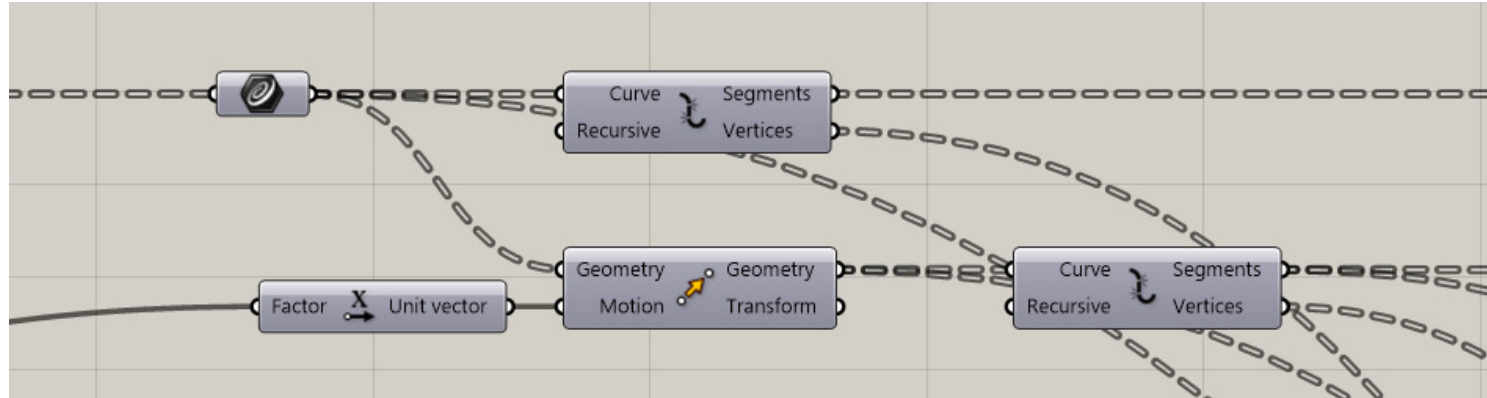
STEP 1 Make the Frames (structure)



The frames are generated using the triamgular component which creates a triangular grid. I then rotated the triangular grid on the YZ plane so I could get the desired form for the “weaving” to occupy.

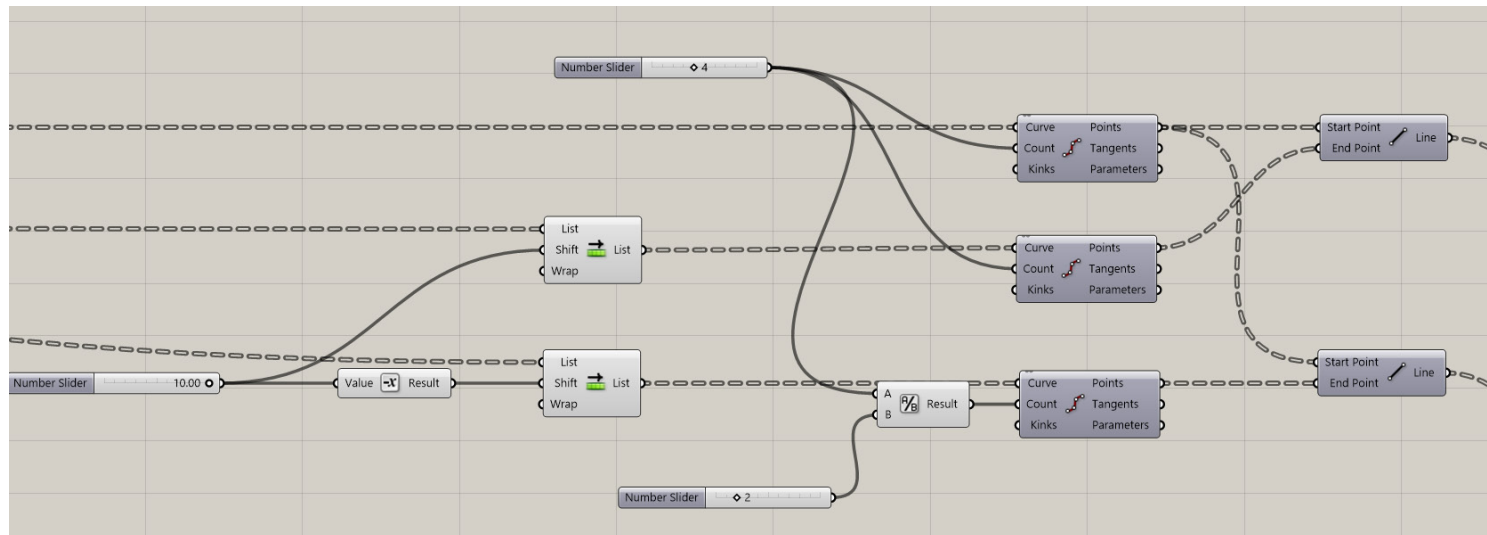
GRASSHOPPER WORKFLOW // FINAL continued

STEP 2 Duplicate + Explode the Frames



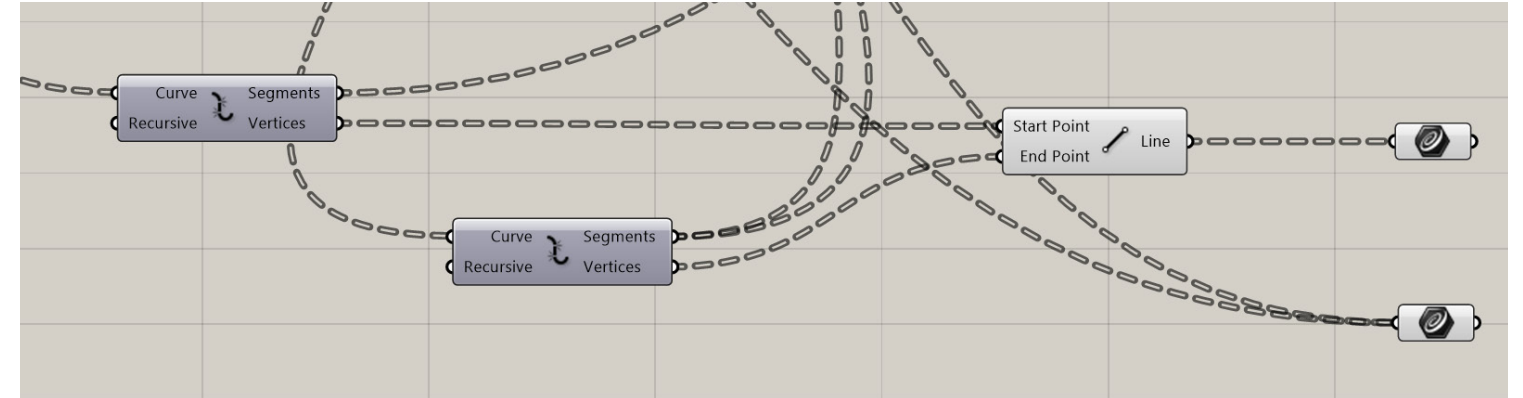
Now that the base geometry is established, the geometry is copied in the x direction and then exploded into individual segments that can be used to connect the threads in the next steps.

STEP 3 Making the Threads



Using the line segments obtained from the previous step, 3 divide curve components are used for each segment of the triangle. For the White threads, the line component is placed with the first set of points being the start point and a second set of points that has been shifted X steps provide the end points of the curve. The black threads are created using the same logic except the list is shifted in the inverse direction for the end points. Because the black threads are treated slightly different from the white threads in the precedent imagery, I felt that dividing the number of points in half on one of the segments resulted in an outcome that was similar to my original script.

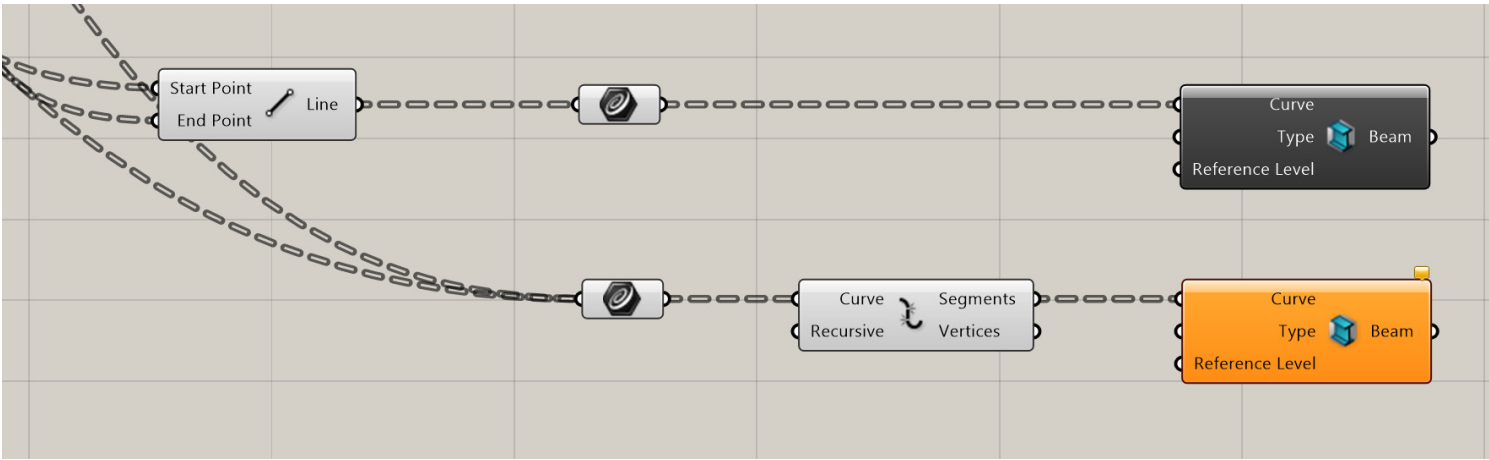
STEP 4 Connect the frames



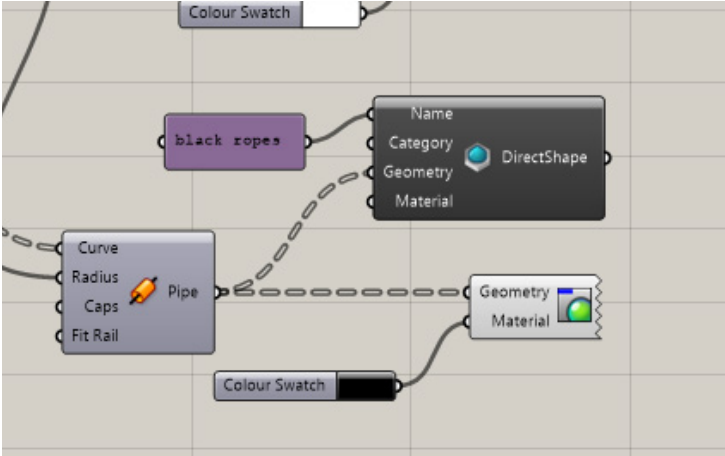
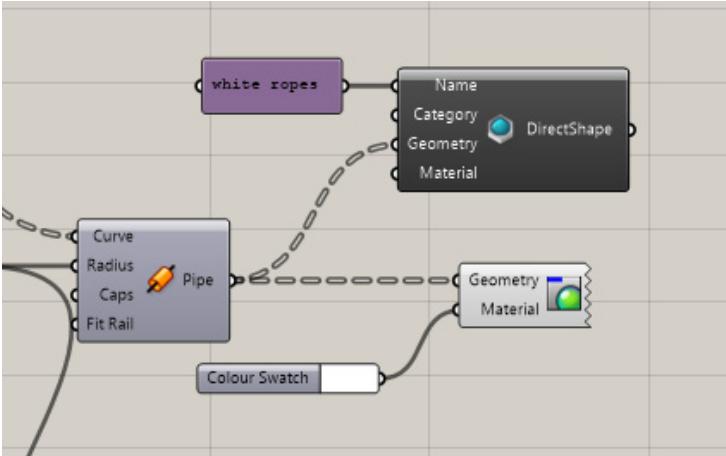
In the final step of the script the frames are connected to each other by using the vertices output as the start and end point of each triangular grid.

RHINO INSIDE

STEP A Adding Structure to the Frame

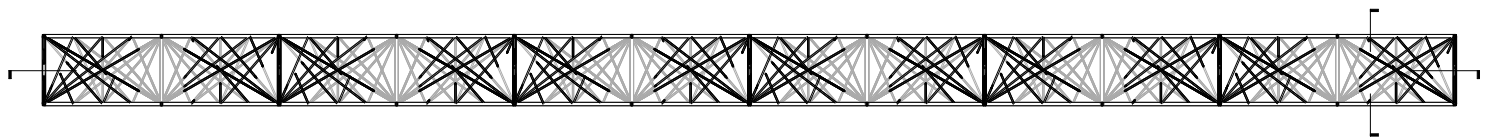


STEP B The Threads

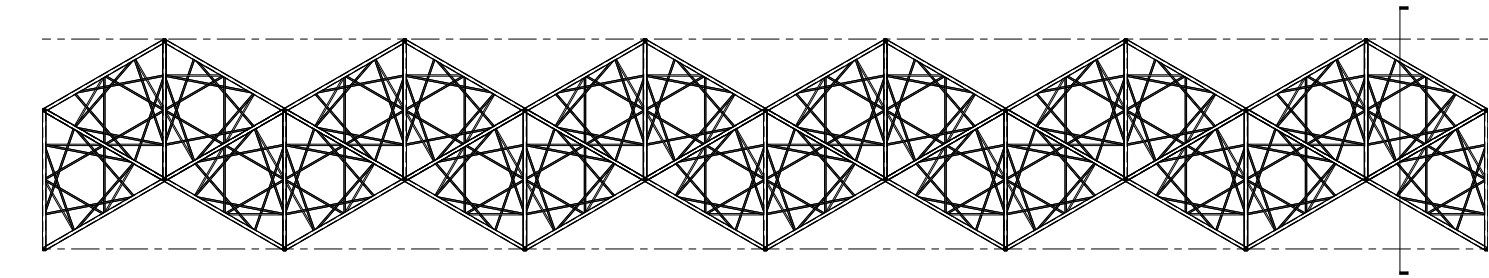


For the threads, I opted to pipe the curves and then add them to Revit using the direct shape component. Because I have 2 of the same conditions, I opted to set the materials for each as different materials in grasshopper so the export to TwinMotion can read these as two separate groups that can have independent materials.

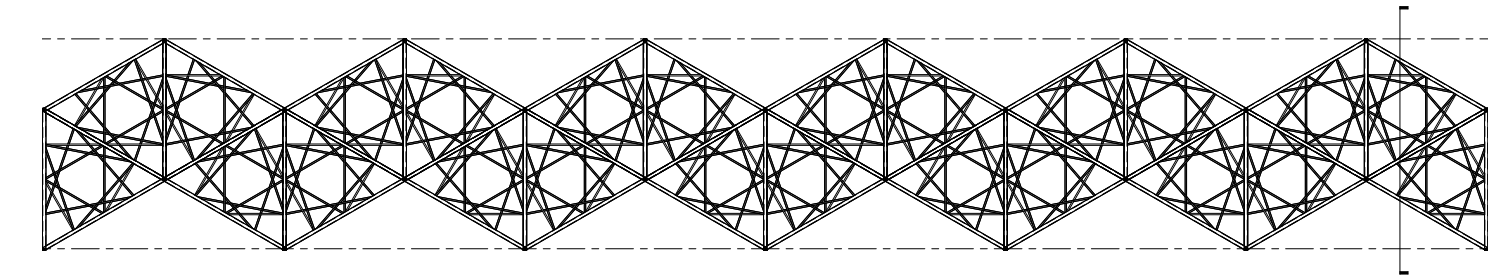
DOCUMENTATION



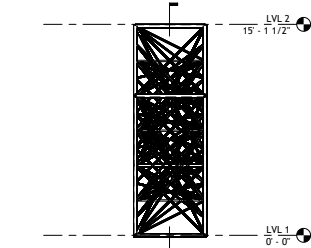
PLAN



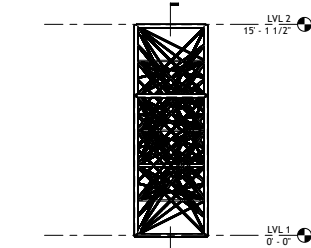
EAST ELEVATION



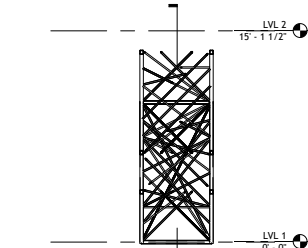
WEST ELEVATION



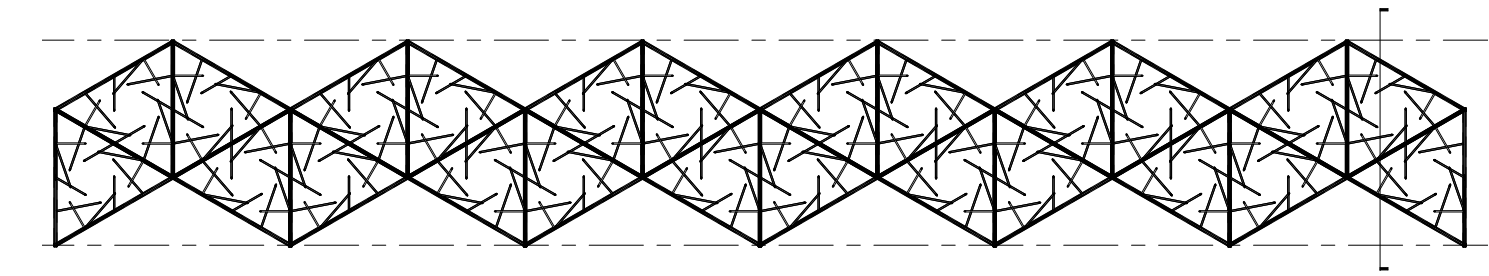
NORTH ELEVATION



SOUTH ELEVATION



SECTION 1



SECTION 2

TWINMOTION IMAGERY

When bringing this project into Twinmotion, I looked at inspiration imagery and found that I was most inspired by the golden hour landscape images. Using that time of day and the warm light, I wanted to create the same feeling in the series of images. I found that dusk during the summer provided the best lighting and I wanted to ensure that the surrounding vegetation was lush and green to support the scene. Despite the design being really simple, I think that using the vegetation as a way to frame the images was successful in the overall compositions.

